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FORM TP 2015189

MAY/JUNE 2015

CARIBBEAN EXAMINATIONS COUNCIL

CARIBBEAN ADVANCED PROFICIENCY EXAMINATION®

ENVIRONMENTAL SCIENCE

ECOLOGY, PEOPLE AND NATURAL RESOURCE USE

UNIT 1 – Paper 02

2 hours 30 minutes

05 MAY 2015 (p.m.)

5189

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions, TWO from each module.
2. Candidates MUST answer ALL questions.
3. Write your answers in the answer booklet provided.
4. You may use a silent, non-programmable, scientific calculator to answer questions.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

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MODULE 1

Answer BOTH questions.

1. (a) Figure 1 shows an ecological pyramid. Study Figure 1 and answer the questions that follow.

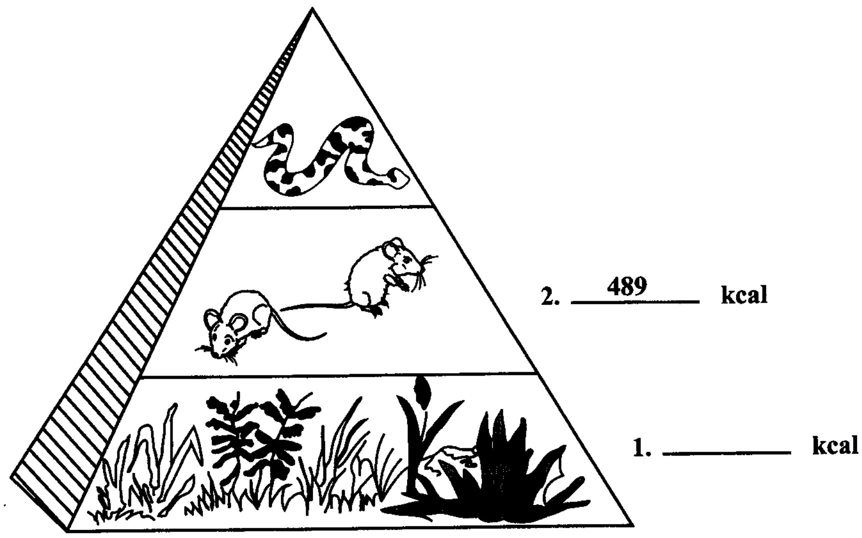

Figure 1. Ecological pyramid

(i) Name the type of ecological pyramid shown in Figure 1. **[1 mark]**

(ii) Use the information in Figure 1 to calculate the amount of energy that is available at the level of the producers. **Show all working.** **[3 marks]**

(b) Outline THREE ways in which the phosphorus cycle may be impacted by the activities of humans. **[6 marks]**

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(c) Figure 2 illustrates the change in population size of iguanas in a mangrove habitat over a 12-year period.

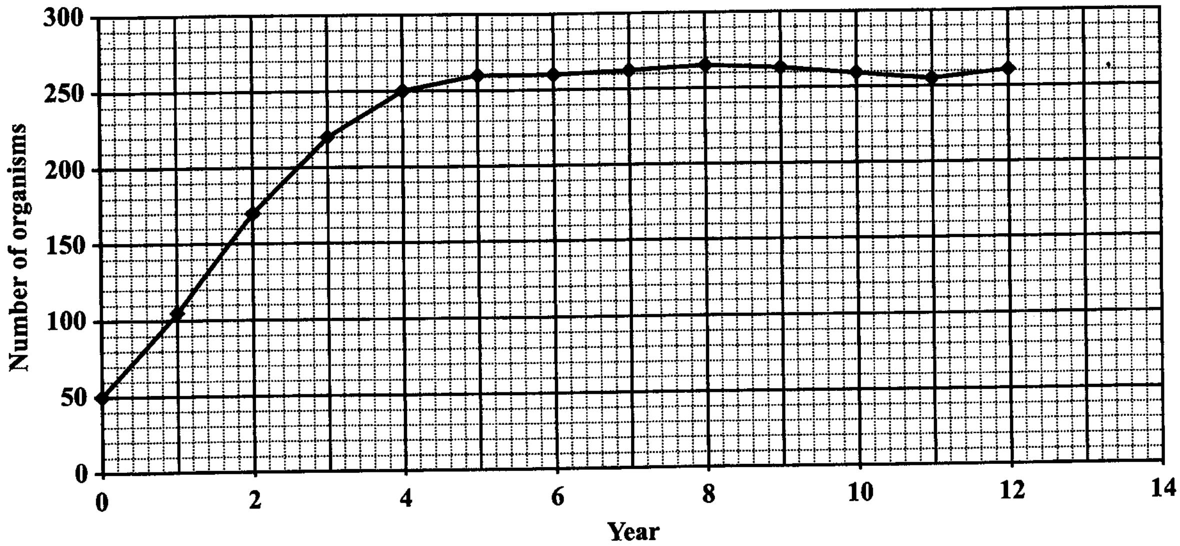

Figure 2. Change in population size of iguanas

(i) What is the population of iguanas in Year 2? **[1 mark]**

(ii) Define the term 'carrying capacity' AND use Figure 2 to estimate the carrying capacity of the mangrove habitat. **[3 marks]**

(iii) Discuss TWO factors that may influence population size at the carrying capacity. **[6 marks]**

Total 20 marks

NOTHING HAS BEEN OMITTED.

2. (a) Outline the role of natural selection in the evolution of a species. **[3 marks]**

(b) Table 1 shows the results from an ecological study using indicator species to investigate water quality in three aquatic environments, A, B and C. Study the information in Table 1 and answer the questions that follow.

TABLE 1: RESULTS OF ECOLOGICAL STUDY

Aquatic Environment	Stoneflies	Beetles	Fly Larvae	Stability Index	Water Quality
A	High	Low	None	10	Good
B	Low	High	Low	7	Intermediate
C	None	Low	High	4	Poor

(i) State FOUR deductions that can be made from the information provided in Table 1. **[4 marks]**

(ii) State which of the three environments is expected to have the GREATEST species diversity AND give ONE reason to support your answer. **[3 marks]**

(iii) Explain, using six points, how species diversity influences ecosystem stability. **[6 marks]**

(c) During the study a student attempted to estimate the population size of manatees in one of the aquatic environments. The data obtained by the student is given in Table 2. Use the information provided in Table 2 to calculate the population size of the manatees.

TABLE 2: RESULTS OF ECOLOGICAL STUDY

Number of manatees caught and marked in first sample	12
Number of marked manatees caught in second sample	5
Total number of manatees caught in the second sample	15

[4 marks]

Total 20 marks

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MODULE 2

Answer BOTH questions.

3. (a) Figure 3 represents the age structure diagram for Country A with a population of 750 000. Study Figure 3 and answer the questions that follow.

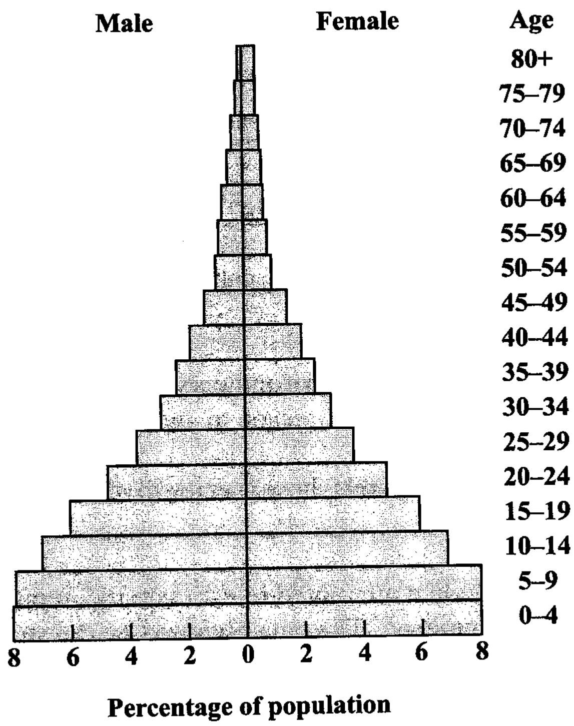

Figure 3. Age structure diagram

(i) Deduce from the diagram whether Country A is a developing or developed country AND explain your answer. **[3 marks]**

(ii) Calculate the number of persons in the population that belong to the pre-reproductive age group. **[4 marks]**

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(b) Explain how temperature influences the distribution AND activities of human populations. **[6 marks]**

(c) Table 3 provides infant mortality estimates for Jamaica, Cuba and Guyana for the year 2012.

TABLE 3: INFANT MORTALITY RATES FOR JAMAICA, CUBA AND GUYANA, 2012

Country	Infant Mortality Rate%
Jamaica	14.30
Cuba	4.83
Guyana	35.59

Source: https://www.cia-gov/library/publications/the-world-factbook/rankorder/209/rant.html

Use the information in Table 3 to explain which country had the FASTEST growing population in 2012. **[4 marks]**

(d) Describe the relationship between per capita GDP and the level of poverty in a country. **[3 marks]**

Total 20 marks

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4. (a) Table 4 presents data on the per capita greenhouse gas emissions for Country B and Country C.

TABLE 4: PER CAPITA GREENHOUSE GAS EMISSIONS

Country	Population Size (millions)	Per Capita Greenhouse Gas Emissions (tonnes of carbon dioxide equivalent per capita)
Country B	9.8	68.6
Country C	2.1	42.0

(i) Define the term 'per capita greenhouse gas emission'. **[2 marks]**

(ii) Explain why people should be concerned about the level of greenhouse gas emissions in the atmosphere. **[3 marks]**

(iii) Identify the country, Country B or Country C, in which greenhouse gases are likely to have the greatest impact AND use the information in Table 4 to justify your answer. **[4 marks]**

(b) Discuss TWO ways by which an increase in the fertility rate of a country may affect a country's ability to achieve sustainable development. **[4 marks]**

(c) Figure 4 shows how the population of a native species and an introduced species in a country changed over a period of seven years. Study Figure 4 and answer the questions that follow.

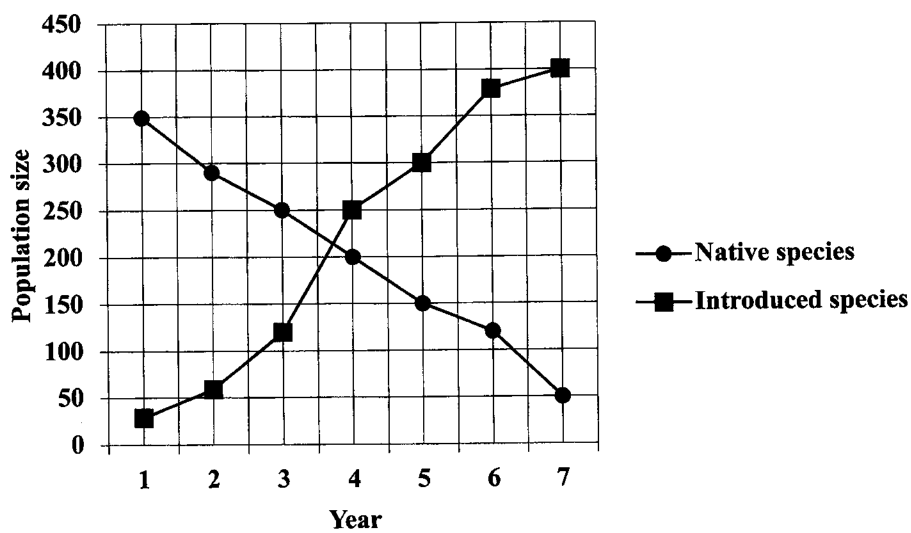

Figure 4. Population of native and introduced species

(i) State the trend in the population of the native species AND the population of introduced species illustrated in Figure 4. **[2 marks]**

(ii) Suggest ONE reason for the trend observed for the native species in Figure 4. **[2 marks]**

(iii) Calculate the average annual decrease in population of the native species for the period represented in Figure 4. **[3 marks]**

Total 20 marks

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MODULE 3

Answer BOTH questions

5. (a) List THREE types of natural resources **other** than beaches that are found in the Caribbean. **[3 marks]**

(b) Figure 5 shows the mean annual beach width for selected beaches in Antigua from 1992 to 1999. Carefully examine Figure 5 and answer the questions that follow.

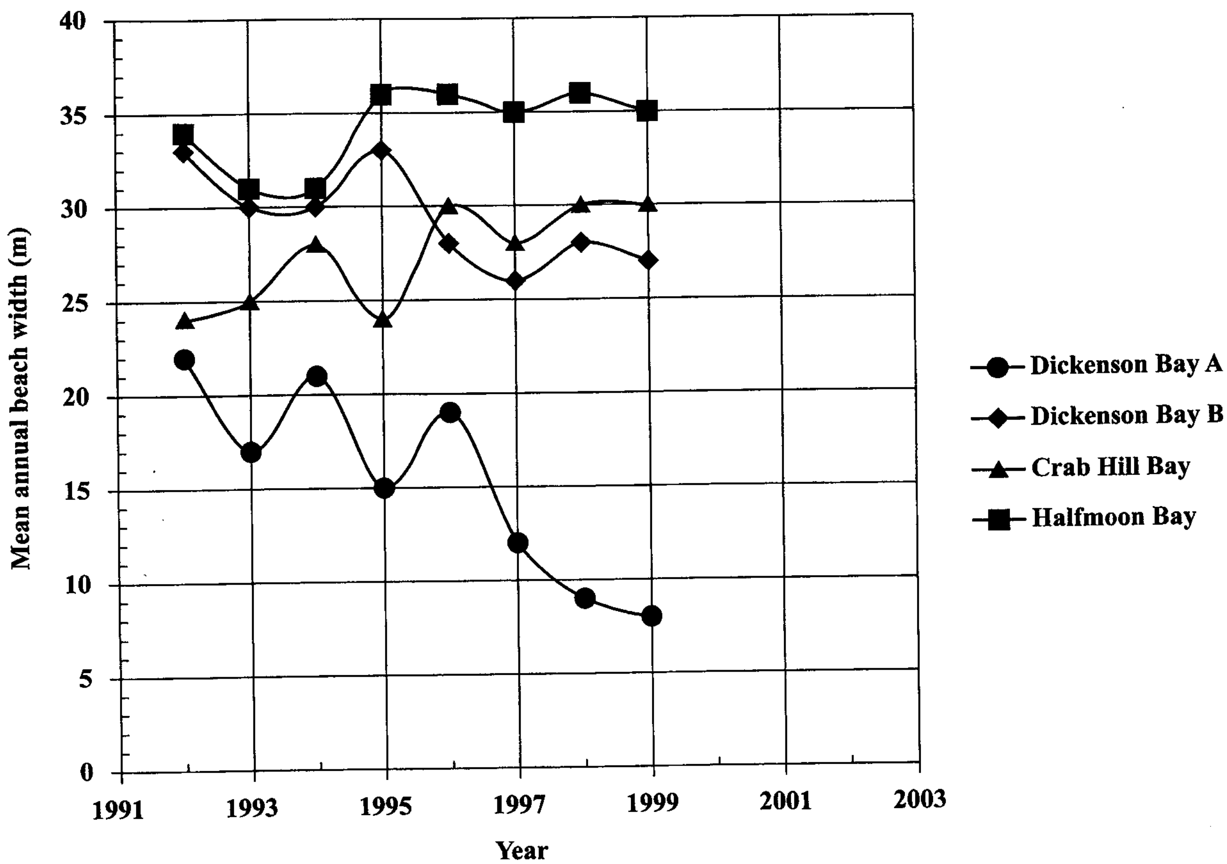

Figure 5. Mean annual beach width for selected beaches in Antigua, 1992–1999

(i) Estimate the average annual rate of change of beach width at Dickenson Bay A over the period 1992 to 1999. Express your answer in m/year and show all working. **[4 marks]**

(ii) Assuming that the trend in beach width at Dickenson Bay A remains the same, predict the width of the beach at Dickenson Bay A in 2002. **[1 mark]**

(iii) Outline FOUR problems that may be caused by the trend of beach width for Dickenson Bay A illustrated in Figure 5. **[8 marks]**

(c) Compare the use of forest resources by indigenous people and commercial operations. In your answer include TWO points. **[4 marks]**

Total 20 marks

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6. (a) Define the term 'non-consumptive' as it relates to the use of natural resources. **[2 marks]**

(b) Figure 6 illustrates the trends in the amount of crude oil transported by sea and the amount of oil spilled by tankers worldwide from 1970 to 2010. Carefully examine Figure 6 and answer the questions that follow.

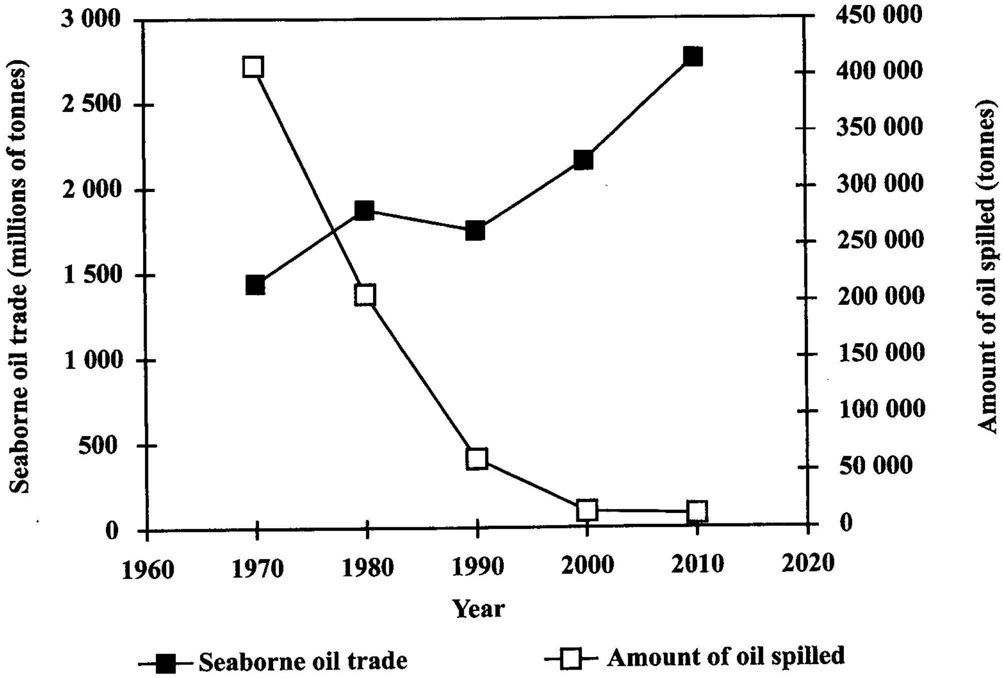

Figure 6. A comparison of the amount of crude oil transported by sea and the amount of oil spilled by tankers, 1970–2010

(i) In which decade did the seaborne oil trade decline? **[1 mark]**

(ii) State the trend in seaborne oil trade AND the trend in the amount of oil spilled as illustrated in Figure 6. **[2 marks]**

(iii) Geograhical and demographic factors could give rise to the trend in the seaborne oil trade while technological factors could give rise to the trend in the amount of oil spilled as illustrated in Figure 6. Explain this statement. **[6 marks]**

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(c) (i) Name the international environmental and conservation agreement which would apply to the activities illustrated in Figure 6. **[1 mark]**

(ii) Outline FOUR principles of the agreement identified in (c) (i) that allow it to be effective. **[4 marks]**

(d) Suggest ONE way in which the activities illustrated in Figure 6 may have an impact on biodiversity AND explain why the effects of this impact are likely to be minimal during the period 1990–2010. **[4 marks]**

Total 20 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.